

GridMind — Conflict & Analysis Test Source

PURPOSE: Synthetic test data for validating source ingestion, project extraction, spatial overlap, temporal overlap, conflict severity, evidence, and AI explanation. NOT a real government document.

Source Title	Satna Utility Coordination Plan — Conflict Test Set
Publisher	Satna Infrastructure Coordination Office (SYNTHETIC TEST DATA)
Source Type	PDF
Publication Date	2026-09-25
Retrieved Date	2026-09-27
Region	Satna, Madhya Pradesh, India
Test Source URL	https://example.com/gridmind/test/conflict-analysis-2026

Utility A — Satna Water & Sewer Utility

Utility Type: WATER | Region: Satna, Madhya Pradesh, India | Website:
https://example.com/gridmind/test/water

Project A1 — Central Civil Lines Water Main Replacement

ID: SAT-WTR-2026-101
Type: WATER_INFRASTRUCTURE
Status: PLANNED
Start: 2026-10-15
End: 2026-12-20
Location: Civil Lines Road, Satna
Geometry: LineString [80.8332,24.5778] → [80.8450,24.5805]
Replace aging water main along the Civil Lines Road corridor. Traffic restrictions may be required.

Project A2 — Civil Lines Road Drainage Upgrade

ID: SAT-WTR-2026-102
Type: DRAINAGE_INFRASTRUCTURE
Status: PLANNED
Start: 2026-11-01
End: 2027-01-10
Location: Civil Lines Road, Satna
Geometry: LineString [80.8390,24.5790] → [80.8510,24.5820]
Upgrade roadside drainage on a corridor overlapping the planned water-main replacement area.

Utility B — Satna Municipal Roads Department

Utility Type: TRANSPORTATION / ROADS | Region: Satna, Madhya Pradesh, India | Website: <https://example.com/gridmind/test/roads>

Project B1 — Civil Lines Road Resurfacing

ID: SAT-ROAD-2026-201
Type: ROAD_CONSTRUCTION
Status: PLANNED
Start: 2026-11-15
End: 2026-12-15
Location: Civil Lines Road, Satna
Geometry: LineString [80.8365,24.5785] → [80.8485,24.5815]
Resurface and partially widen Civil Lines Road during the same period as water and drainage projects.

Project B2 — Station Road Street Lighting

ID: SAT-ROAD-2026-202
Type: STREET_LIGHTING
Status: PLANNED
Start: 2027-02-01
End: 2027-03-15
Location: Station Road, Satna
Geometry: LineString [80.8200,24.5700] → [80.8260,24.5720]
Install street-lighting infrastructure. Intentionally outside the Civil Lines overlap window.

Expected Conflict Test Cases

Case	Projects	Expected Result
1	A1 ↔ B1	HIGH / strong spatial + temporal overlap candidate
2	A2 ↔ B1	HIGH or MEDIUM / spatial + temporal overlap candidate
3	A1 ↔ A2	Spatial + temporal overlap; verify same-utility comparison rule
4	A1 ↔ B2	NO TEMPORAL CONFLICT
5	A2 ↔ B2	NO TEMPORAL CONFLICT

Validation Checklist

1. Extract source metadata and preserve URL/sourceId.
2. Extract all 4 projects with IDs, utility, type, dates, location and geometry.
3. Interpret coordinates as [longitude, latitude].
4. Calculate date overlap deterministically.
5. Calculate spatial intersection/nearby corridor deterministically.
6. Link conflicts to source/project evidence.
7. AI explanation uses only calculated facts and evidence.
8. Unsupported facts are UNKNOWN, never invented.
9. Analysis run reaches COMPLETED/PARTIAL/FAILED correctly.